

Bridging the Reality Gap: Immersive VR Simulation for Enhancing Stress Resilience in Medical Students

Sebastian Weiss* Laura Hanke† Erik Schröder Marina Weinberg Theresa Dicks
 Christoph von Castell Marlene Wessels Heiko Hecht Pavel Dietz Tobias Huber
 Christian Hansen



Figure 1: Example scenario location and NPC: Supervisor of residents in their office.

ABSTRACT

Medical students often face a “reality gap” between theory and clinical practice, leading to burnout. We present a system to induce stress as first step for later stress resilience trainings through immersive simulations of clinical conflicts. Based on a survey of 374 professionals, we developed four scenarios targeting stressors like “conflict with superiors” and “time pressure,” utilizing realistic locations and NPCs. Our framework uses a Meta Quest 3 and a custom Android app for real-time experimenter control. To overcome static training limitations, we explore local Large Language Models (LLMs) for generative NPC behavior. We detail a work-in-progress evaluation protocol using physiological measures (HR, HRV, BF) and questionnaires, providing a scalable framework for medical stress inoculation training.

Index Terms: Virtual reality, Stress, Resilience, Medical studies, Training, Education.

1 INTRODUCTION

While medical students acquire vast theoretical knowledge, the gap between education and high-pressure clinical reality—such as emergency rooms—presents significant challenges. Medical students and early-career physicians are particularly susceptible to high levels of stress, which can impair their mental health, negatively impact learning outcomes, and diminish long-term professional engagement. Preliminary research indicates that specific stressors are perceived as particularly burdensome, including conflicts with superiors and colleagues, working under time pressure, fear of making treatment errors, interruptions, and a lack of information when making decisions. Furthermore, the anticipation of this burden significantly impacts career choices; for instance, a

shortage of young talent in surgery is evident, partly due to students avoiding the field based on the anticipated workload and stress. Current national competency-based learning catalogs address stress management only rudimentarily, typically late in the curriculum (5th–6th semester) [19]. Consequently, there are no established structured courses for “coping strategies” to help young professionals navigate critical situations, such as breaking bad news or managing complex trauma cases under time constraints. This lack of preparation creates a discrepancy between expectation and reality. This anxiety and the fear of errors can paralyze decision-making processes. Without a “mental health toolkit” (see e.g. [3]) or structured coping mechanisms, the cognitive and psychological demands of the job can lead to burnout, endangering the practitioner’s long-term health and, by extension, patient safety.

A stressful work environment is comprised of a multitude of factors like interpersonal and interprofessional communication, multitasking, and time pressure, which are often aggravated by basic needs like hunger or sleep deprivation. Prior research shows that experienced physicians were able to navigate these encumbrances more easily than novices [13, 22], which suggests that stress management is a learnable and thus a teachable skill. Simulation in medical education has been proven to be a valuable asset in the acquisition of technical and communicative skills [16, 10], but simulation for multitasking and stress resistance in surgical education is a novel concept with limited prior research [9]. The inability to access skills during critical moments due to stress highlights the need for training that goes beyond textbook knowledge to include resilience and stress management.

To address these deficits, virtual reality (VR) has emerged as a promising solution for training stress resilience. VR technology allows for the creation of immersive, computer-generated worlds with a high degree of realism, capable of simulating the complex sensory environments of an operating theater or a ward. The primary advantage of VR-based training is the ability to reproduce stressful scenarios—such as interpersonal conflicts or time-critical decisions—in a safe, repeatable, and controlled environment.

VR is considered efficient because it enables “training for reality” without risk to the patient. It allows students to experience

*e-mail: weiss@isg.cs.uni-magdeburg.de

†e-mail: Laura.Hanke@unimedizin-mainz.de

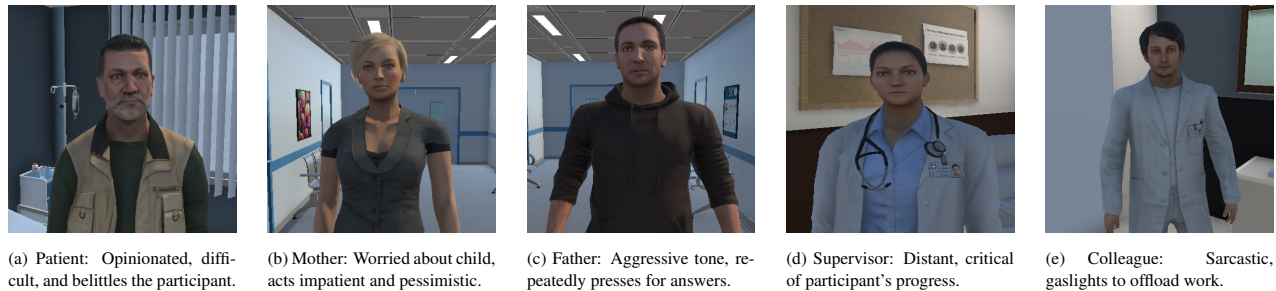


Figure 2: NPCs from the four clinical scenarios, taken from Microsoft's Rocketbox.

high-load situations, such as managing a difficult conversation with a senior physician or handling demanding relatives under time pressure, while simultaneously performing cognitive tasks. This immersion facilitates the application and testing of coping strategies that might otherwise be theoretical. By combining VR simulations with preparatory e-learning focused on stress management resources, students can build a "mental health toolkit" and immediately apply it in simulated high-stress contexts. This approach aims to transform abstract coping mechanisms into practical, retrievable skills.

Building upon the identified need for improved stress management training, this paper explores the development of a VR scenario which will later be compared to e-learning materials. The overarching research interest lies in determining whether a training program that realistically illustrates the stress-inducing aspects of everyday medical work can reduce cognitive and emotional loads in typical clinical situations.

The project this research is embedded in has the following research goals:

1. creation of an efficient e-learning to teach coping skills specific to physicians.
2. creation of a multi-level, high stress environment in VR.
3. Testing the effect of the e-learning in a randomized controlled study using the VR environment.

Working towards these goals, this WIP presents a technical foundation for stress induction rather than a validated curriculum. Specifically, we deliver four primary contributions:

- **System Infrastructure:** A dual-component architecture combining a standalone VR application for trainees with a tablet-based interface for real-time experimenter control and monitoring.
- **Scenario Design:** A set of four clinical scenarios targeting social stressors (e.g., conflict with superiors, time pressure) grounded in a survey of $n=374$ medical professionals.
- **Evaluation Protocol:** A multimodal study design combining objective physiological measurements (HR, HRV, BF) with subjective instruments (NASA-TLX, SAM) to validate stress induction capabilities.
- **Local LLM Feasibility:** An early-stage exploration of on-device LLMs (specifically Llama 3.1 8b) to assess the technical viability of generative, non-scripted NPCs in standalone VR headsets. Future work will focus on the empirical validation of these scenarios for stress induction and the subsequent longitudinal assessment of resilience acquisition.

The following sections will detail the methodology used to create the VR software that allows participants to experience these scenarios, remote control software to operate the program, and the experimental design employed to measure their efficacy in terms of stress induction.

2 RELATED WORK

The role of virtual reality (VR) in medical education has evolved from training isolated technical skills to simulating complex, high-stakes clinical environments [5]. Our work lies at the intersection

of immersive medical simulation, social stress induction, and generative non-player characters (NPCs).

Simulation-based training has traditionally relied on high-fidelity mannequins to rehearse acute medical crises [5]. VR provides a scalable alternative that can evoke comparable physiological responses without the logistical demands of simulation centers [17]. While VR-based technical skill trainers are well established, research on multitasking and stress resilience remains limited [9]. Early studies suggest that VR can reproduce the cognitive and emotional demands of emergency situations [15], yet most systems depend on scripted interactions, restricting realism and unpredictability.

Stress in clinical practice emerges from interpersonal communication, multitasking, and time pressure [20]. Prior VR training for nurses [21], EMTs [17], and firefighters [1] demonstrates that virtual stressors can induce significant physiological arousal. In medical education, social stressors such as conflicts with superiors or interactions with relatives are perceived as particularly challenging by students [19]. Weiß et al. showed that virtual social conflicts can effectively reduce the perceived gap between training and real clinical practice [20].

Recent advances in Large Language Models (LLMs) enable a shift from scripted scenarios toward generative VR simulations [8]. LLM-driven NPCs can respond contextually and emotionally, enhancing immersion and social realism. Building on this, our work evaluates storage and performance trade-offs of local LLMs (e.g., Llama 3.1) to support portable, infrastructure-light VR training suitable for medical education.

3 MATERIALS AND METHODS

To create software that can be used as a research tool for our controllable immersive training environment, we designed a dual-component architecture consisting of a VR application for trainees and a mobile tablet application for experimenter-led moderation. This methodology focuses on the creation of high-fidelity social scenarios that allow medical students to practice stress-inoculation strategies without risking patient safety.

3.1 Scenario Creation

Our scenarios are based on a preliminary empirical survey of $n=374$ medical professionals (281f, 93m, 2d; mean age=34.72). Participants rated situational stressors on a 1–10 (10 being very stressful) scale, identifying **Time Pressure** ($\phi=7.3$), **Conflict with Superiors** ($\phi=6.78$), and **Colleague Conflicts** ($\phi=6.52$) as the most burdensome. Other important stressors have been identified as well: **Being afraid to make a mistake in patient treatment** ($\phi=6.5$); **Insufficient information when making decisions** ($\phi=6.41$); **Interruptions** ($\phi=6.29$), and **Lack of support** ($\phi=6.24$). Notably, medical students reported significantly higher expected stress levels than the actual stress perceived by junior doctors ($p<0.05$). Consequently, we developed four scenarios with medical experts, each targeting these identified stressors. In addition, mental load tasks were implemented. Table 1 provides a compact overview of tar-

geted stressor, key events, duration and experimenter controls for each scenario.

In addition to the scenarios, we created two mental load tasks for the participants: the n-back test [14] and mental arithmetic [18]. We collect information on the number of tasks solved and errors made during task completion. These tasks have a difficulty setting, where with higher difficulty we decrease the available time to solve the tasks.

3.2 VR System

To ensure controlled exposure to the scenarios, we developed a software system consisting of a VR application running on a Meta Quest 3 HMD and a remote-control application for Android tablets. Both applications were implemented in Unity v6 (v6000.0.60f1)¹ using C#. Physiological data are recorded on a PC via Bluetooth, while experimental logs are stored on the HMD and tablet. After each session, the tablet retrieves the log files from the HMD for transfer to a PC and subsequent analysis. Figure 3 illustrates the system architecture.

3.2.1 Headset Application

The VR application comprises eight scenes: a lobby, four scenario scenes, two mental workload tasks (a math task and an n-back test), and a questionnaire scene. The latter includes NASA-TLX [11], SAM valence and arousal ratings [4], and assessments of visually induced motion sickness and eye strain using the FMS and FMS-O questionnaires [12].

Scenario scenes feature NPCs representing scenario-specific roles (cf. Figure 2). Character models are sourced from Microsoft's Rocketbox package [7]² and animated using pre-recorded idle, talking, and yelling animations from Adobe Mixamo³. Meta's Lipsync SDK provides viseme synchronization. Each NPC is assigned multiple pre-recorded voice lines to enhance social realism.

3.2.2 Tablet Application

The VR and tablet applications communicate via TCP/IP with automatic discovery and require a shared WiFi network. The tablet interface allows the experimenter to control scene transitions, NPC voice lines, scenario events, and task difficulty for both cognitive tasks. To assist participants when needed, the VR application

streams the participant's first-person view to the tablet (see Figure 4).

After each experimental run, the tablet downloads log files from the HMD for collection and statistical analysis on a PC or Mac using IBM SPSS v28⁴.

3.3 Stress Measures

To make assumptions about the stress-inducing capabilities of the scenarios, we collect several physiological measures, i.e. heart rate (HR), heart rate variability (HRV), and pulse (1/min) using the integrated, chest-worn "Zephyr BioHarness BTLE ECHO"⁵ (Medtronic, Boulder, CO). These measures have been used in related work and have been found to be adequate for this kind of stress-related research [6]. The device samples 1 channel ECG at 1kHz and provides this information at a frequency of 250 Hz, and Breathing Frequency (BF) at a sampling rate of 1Hz. Data is transmitted via Bluetooth to the recording PC and synchronized with the VR application timestamps. To establish a physiological baseline, participants rest for 5 minutes (acclimation phase) followed by a 3-minute baseline recording before the first scenario. Raw Inter-Beat Intervals (IBIs) are extracted from the ECG signal. The device applies filtering already to ensure data quality. Segments containing significant speech artifacts are flagged, as vocalization significantly alters vagal tone. We calculate the Root Mean Square of Successive Differences (RMSSD) as our primary time-domain measure of Heart Rate Variability (HRV). RMSSD was selected because it reflects parasympathetic nervous system activity and is relatively robust to respiratory influences compared to frequency-domain measures in short-duration recordings. Breathing Frequency (BF) is monitored concurrently to interpret HRV changes. Specifically, BF is used to control for Respiratory Sinus Arrhythmia (RSA), ensuring that observed changes in HRV are attributable to the induced social stress rather than changes in respiration rate.

3.3.1 Other Measures

Before the first and after each scenario, participants answer the NASA-TLX and FMS(O) and further rate their arousal and valence via the SAM scale.

3.4 Study Procedure

Participants will be recruited via mailing lists, public notices, announcements in class, and social media channels. Participants are required to be at least 18 years of age and enrolled in the 5th semester of human medicine studies. People with impaired vision or hearing are excluded.

3.4.1 Experiment Protocol

The planned procedure for this experiment, including the herein described materials, has been approved by the ethics board of the Institute for Psychology, Johannes Gutenberg-University Mainz. After arrival, participants will be required to sign informed consent. A research assistant will adjust the sensor (see 3.3), whereafter they will introduce to the Meta Quest 3 HMD. Participants will experience scenarios in a counterbalanced order. Each scenario scene is followed by the questionnaire scene (see 3.3.1). Physiological data will be collected, and changes will be calculated as percentages; a deviation of > 10 % from the baseline will be considered a reaction. The experiment is planned as a within-subject study to test the efficacy of stressors. In the overarching project, the trial data will be used for further comparison to investigate whether access to e-learning programs is more effective before (intervention) or after (control) the VR scenarios. Piloting has shown that the experiment

¹www.unity.com, last opened December 28th, 2025
²https://github.com/microsoft/Microsoft-Rocketbox, last opened December 28th, 2025
³www.mixamo.com, last opened December 28th, 2025

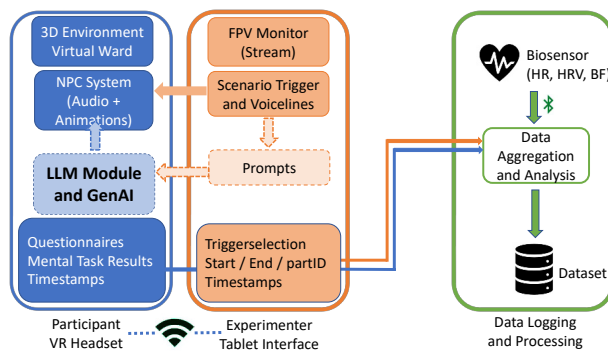


Figure 3: Systematic overview of system parts and their relationships. Dashed modules are experimental.

⁴https://www.ibm.com/docs/en/spss-statistics/28.0.0, last opened December 29th, 2025

⁵https://www.zephyranywhere.com, last opened December 29th, 2025

Table 1: Overview of Clinical Scenarios, Stressors, and Controls

No.	Target Stressor	Key Events & Mechanism	Est. Duration	Experimenter Controls
1.	Conflict with patients; Time-sensitive decision making	Patient arrives for elective surgery planning; exhibits “Dr. Google” behavior; demands higher hierarchy.	~5min.	Trigger voice lines; Adjust aggression.
2.	Interruptions; Aggressive tone	Participant intercepted by parents; parents insist on answers, interrupt repeatedly, and exhibit pessimism.	~5min.	Control interruption frequency; Trigger questions.
3.	Conflict with superiors; Criticism	Meeting with supervisor; distant/objective tone; performance criticized as “minimum requirement”; demands commitment.	~4min.	Control tone; Advance dialogue.
4.	Lack of support; Sarcasm	Colleague pressures participant for paper-work/shift cover; uses snarky/gaslighting tone.	~3min.	Trigger sarcastic comments; Request favors.

takes about 45 to 60 min from the participant’s arrival to exiting the lab.

3.5 Future Directions

Looking forward, we want to continue developing this software and make use of generative AI (GenAI) for improved NPC “acting”. This approach has several benefits. First, we assume that with increased resilience, the effect of the stressors diminishes. We further assume that repetitive use of static scenarios decreases their efficacy. Thus, we intend to create a system that slightly modifies interactions of NPCs based on their roles and relationships with one another and the VR user, respectively. Secondly, using real-time biofeedback, it may be possible to add further stressors into the scenario to increase the stress, thereby increasing demand on the VR user and further aiding them in building resilience.

Table 2: Comparison of local LLMs (Transposed).

Metric	Llama 3.1 8b	Mistral 7b	GPT-OSS 20b	Zephyr 7b	Qwen3 8b
Params (bn)	8	7	20	7	8
Storage (GB)	4.9	4.4	13.0	4.1	5.2
Speed (tok/s)	131.7	142.1	37.5	149.7	123.1
Context (tok)	128k	32k	128k	32k	40k
Index	19	1	49	1*	25

3.5.1 GenAI Integration

An integration of Gen-AI necessitates three components: audio transcription to “hear” what participants say, text generation to formulate an answer, and text-to-speech (TTS) to turn the answer into an audible response. To be run on-device and thus at any location, the text generation and the TTS components require their own trained model, have a high output speed, and come with little storage requirement. Additionally, the model must be able to create text in German language that is meaningful and convincing in terms of content. Table 2 compares a first selection of openly available models.

We created a simple priming prompt for the LLMs to play the patient in the first scenario to compare the quality of the conversations. We prompted each model repeatedly with “Hello” and “Goodbye.” The best results were achieved with Llama 3.1 8b, whereas other models gave repetitive answers, which is probably due to the repetitive prompts.

3.5.2 Prompt Engineering

To instruct an LLM to mimic a given persona, we improved the prompt from the previous section. Following guidelines from [2] we created comprehensive prompts including a basic ruleset, a role description for the LLM, a role description for the VR user, wanted tonality, a description of the locale and context of the scene, and lastly, further instructions for the LLM to stay within the storyline when creating replies. During the prompt engineering process, we observed a few anomalies, such as the introduction of logical and grammatical errors or an inability of the LLM to adhere to strict limitation rules, i.e., limiting the number of sentences was not followed exactly. Furthermore, we found that the strength of tone varies between individual executions of a scenario. Further prompt engineering is required for satisfactory results, but this work-in-progress is very promising for context-stable yet flexible scenarios instead of the current static scenarios.

4 CONCLUSION

In this paper, we presented the design and realization of a technical framework aimed at medical stress inoculation training. By combining a Meta Quest 3 application with a mobile remote-control interface, we have created a controllable environment for simulating critical clinical social interactions. Our preliminary investigation into local LLMs, specifically Llama 3.1 8b, indicates that generative NPCs offer a viable solution to the limitations of repetitive, static training scenarios. While a formal user study is required to validate the stress-inducing efficacy of these scenarios through physiological measures like HR and HRV, this work provides a scalable foundation for bridging the gap between theoretical medical education and the high-pressure reality of clinical practice.



Figure 4: Screenshot of tablet app. Highlighted areas show grouped functionalities. Green: ParticipantID and scenario. Red: Difficulty. Blue: VR FPV. Purple: NPC triggers for scenarios. Orange: Miscellaneous settings.

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